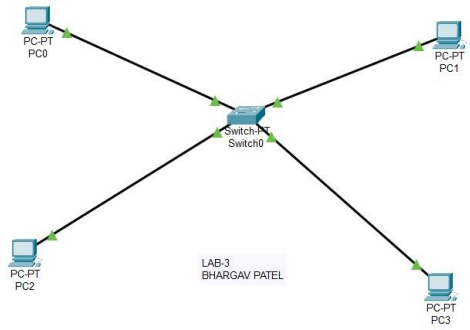
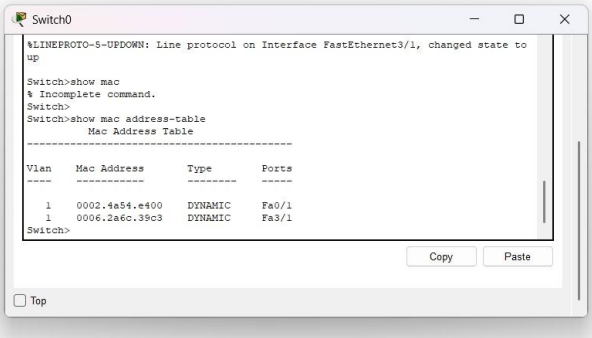
LAB-3

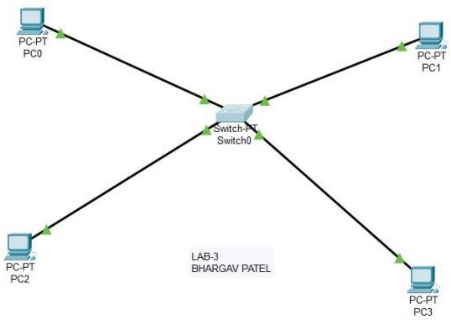
Name: Bhargav Patel

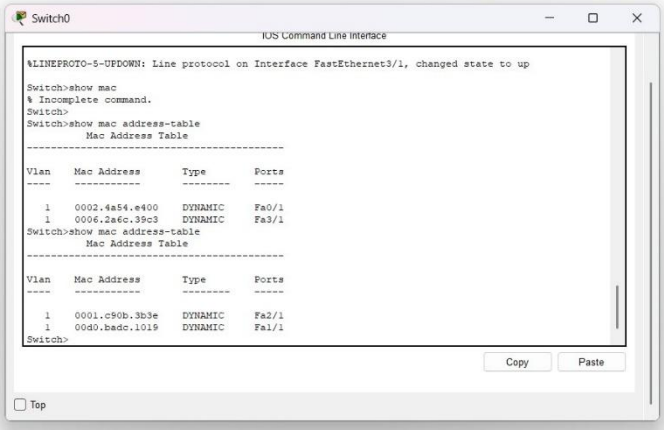
Banner ID: 916503575

❖ LAB 3

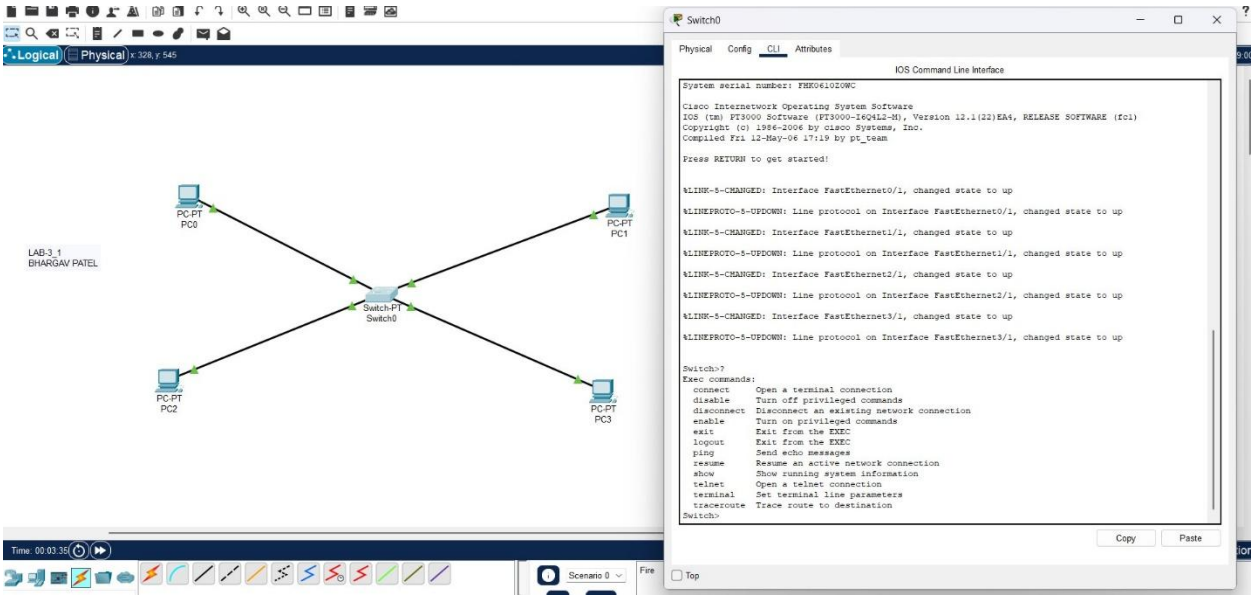


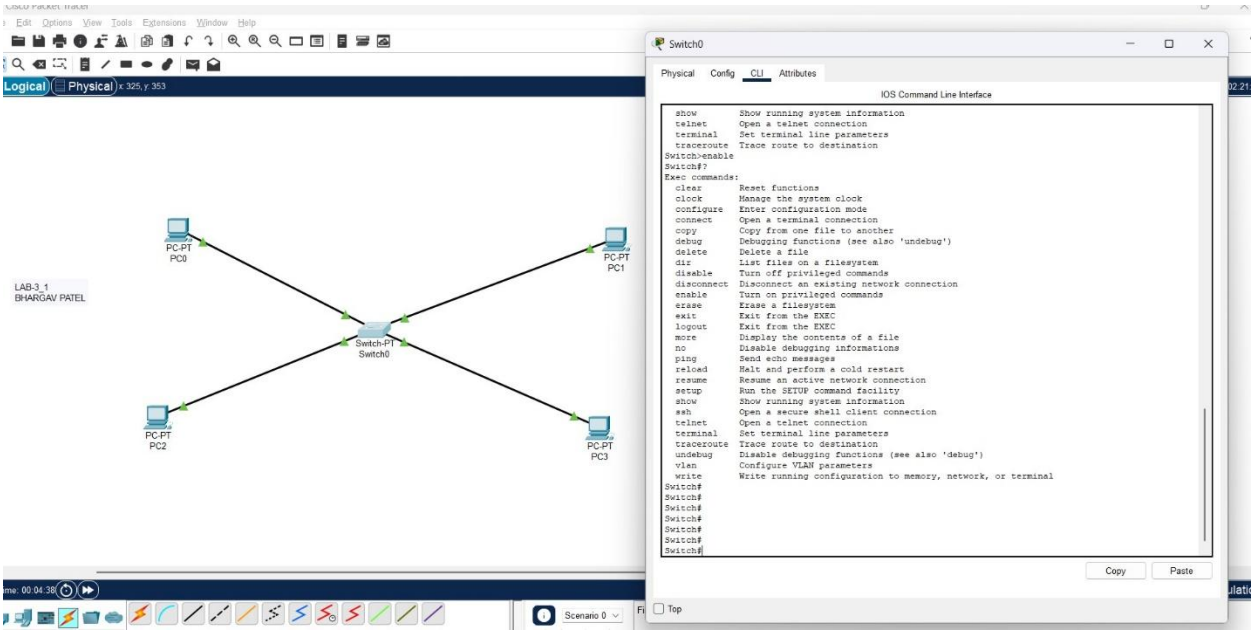




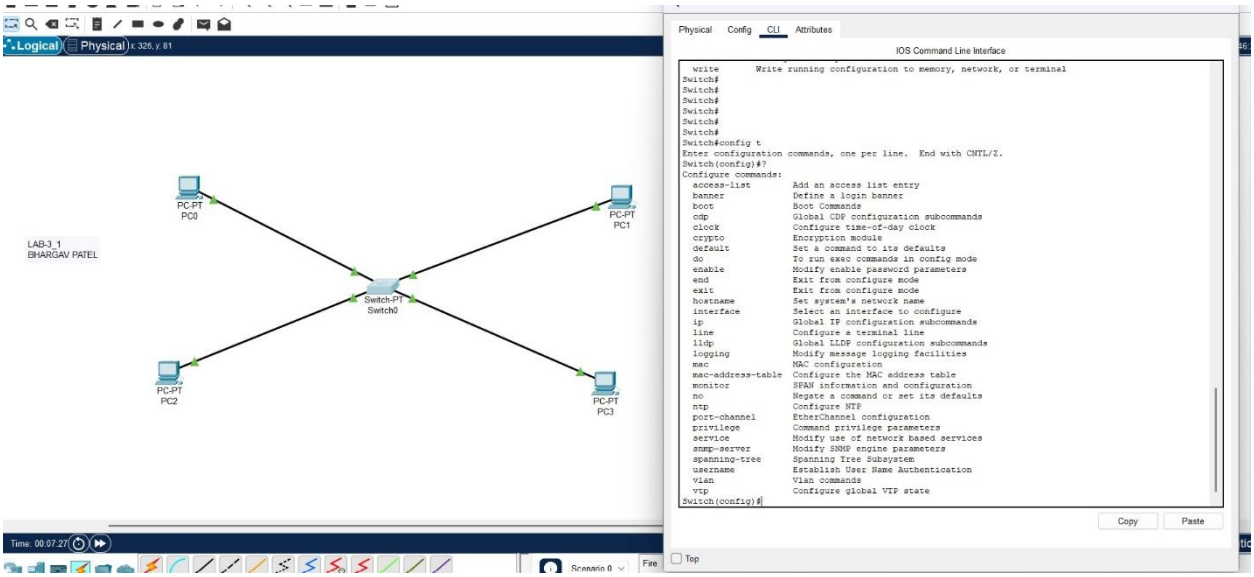


LAB-3


Logical
Physical
LAB-3_1
BHARGAV PATEL
PC-PT PC0
PC-PT PC1
PC-PT PC2
PC-PT PC3
Switch-PT Switch0
Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
System serial number: FHK0610Z0WC
Cisco Internetwork Operating System Software
IOS (tm) PT3000 Software (PT3000-I6Q4L2-M), Version 12.1(22)EA4, RELEASE SOFTWARE (fc1)
Copyright (c) 1986-2006 by cisco Systems, Inc.
Compiled Fri 12-May-06 17:19 by pt_team
Press RETURN to get started!
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet2/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet3/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet3/1, changed state to up
Switch>?
Exec commands:
connect Open a terminal connection
disable Turn off privileged commands
disconnect Disconnect an existing network connection
enable Turn on privileged commands
exit Exit from the EXEC
logout Exit from the EXEC
ping Send echo messages
resume Resume an active network connection
show Show running system information
telnet Open a telnet connection
terminal Set terminal line parameters
traceroute Trace route to destination
Switch>
Copy
Paste
Top


Logical
Physical
LAB-3_1
BHARGAV PATEL
PC-PT PC0
PC-PT PC1
PC-PT PC2
PC-PT PC3
Switch-PT Switch0
Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
show Show running system information
telnet Open a telnet connection
terminal Set terminal line parameters
traceroute Trace route to destination
Switch>enable
Switch#?
Exec commands:
clear Reset functions
clock Manage the system clock
configure Enter configuration mode
connect Open a terminal connection
copy Copy from one file to another
debug Debugging functions (see also 'undebug')
delete Delete a file
dir List files on a filesystem
disable Turn off privileged commands
disconnect Disconnect an existing network connection
enable Turn on privileged commands
erase Erase a filesystem
exit Exit from the EXEC
logout Exit from the EXEC
more Display the contents of a file
no Disable debugging informations
ping Send echo messages
reload Halt and perform a cold restart
resume Resume an active network connection
setup Run the SETUP command facility
show Show running system information
ssh Open a secure shell client connection
telnet Open a telnet connection
terminal Set terminal line parameters
traceroute Trace route to destination
undebug Disable debugging functions (see also 'debug')
vlan Configure VLAN parameters
write Write running configuration to memory, network, or terminal
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Copy
Paste
Top

LAB-3

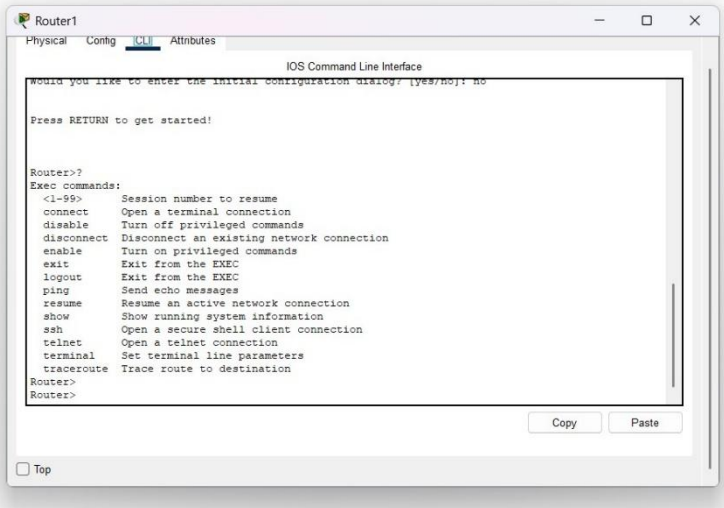

Logical
Physical
LAB-3_1
BHARGAV PATEL
PC-PT
PC0
PC-PT
PC1
Switch-PT
Switch0
PC-PT
PC2
PC-PT
PC3
Time: 00:07:27
Physical
Config
CLI
Attributes
IOS Command Line Interface
write Write running configuration to memory, network, or terminal
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#?
Configure commands:
access-list Add an access list entry
banner Define a login banner
boot Boot Commands
cdp Global CDP configuration subcommands
clock Configure time-of-day clock
crypto Encryption module
default Set a command to its defaults
do To run exec commands in config mode
enable Modify enable password parameters
end Exit from configure mode
exit Exit from configure mode
hostname Set system's network name
interface Select an interface to configure
ip Global IP configuration subcommands
line Configure a terminal line
lldp Global LLDP configuration subcommands
logging Modify message logging facilities
mac MAC configuration
mac-address-table Configure the MAC address table
monitor SPAN information and configuration
no Negate a command or set its defaults
ntp Configure NTP
port-channel EtherChannel configuration
privilege Command privilege parameters
service Modify use of network based services
snmp-server Modify SNMP engine parameters
spanning-tree Spanning Tree Subsystem
username Establish User Name Authentication
vlan Vlan commands
vtp Configure global VTP state
Switch(config)#
Copy
Paste
Top

LAB-3

❖ LAB 3_1

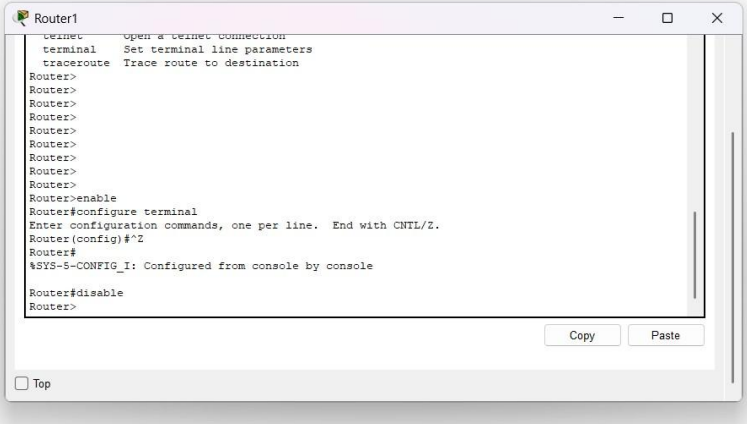
LAB-3_1
BHARGAV PATEL



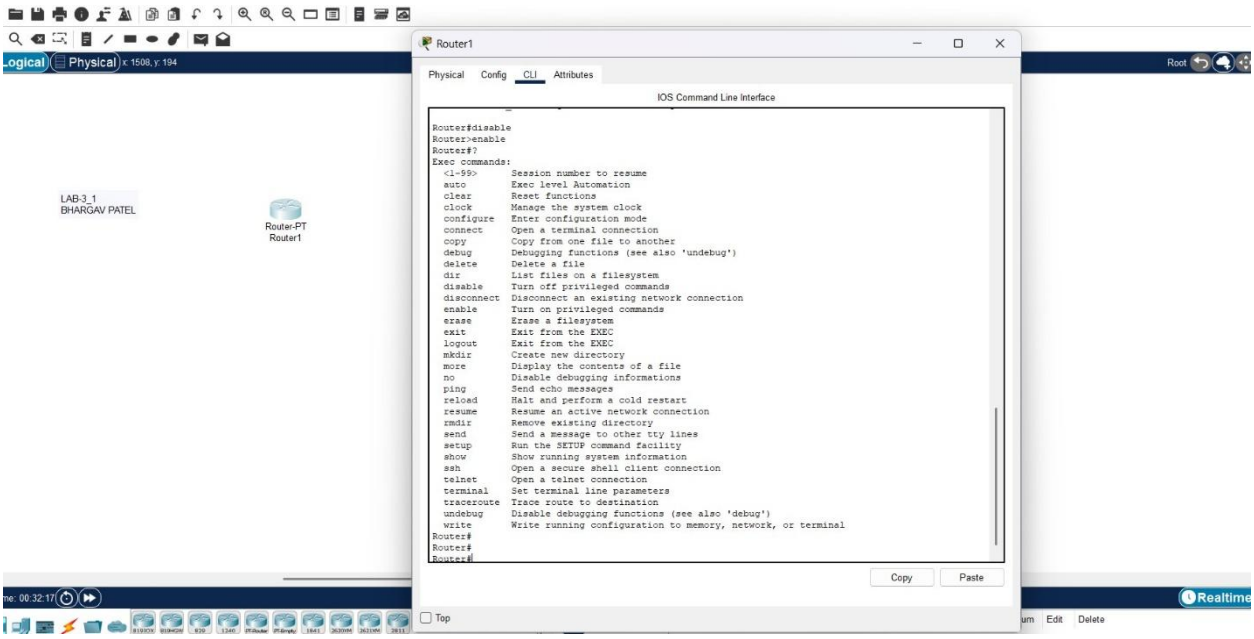


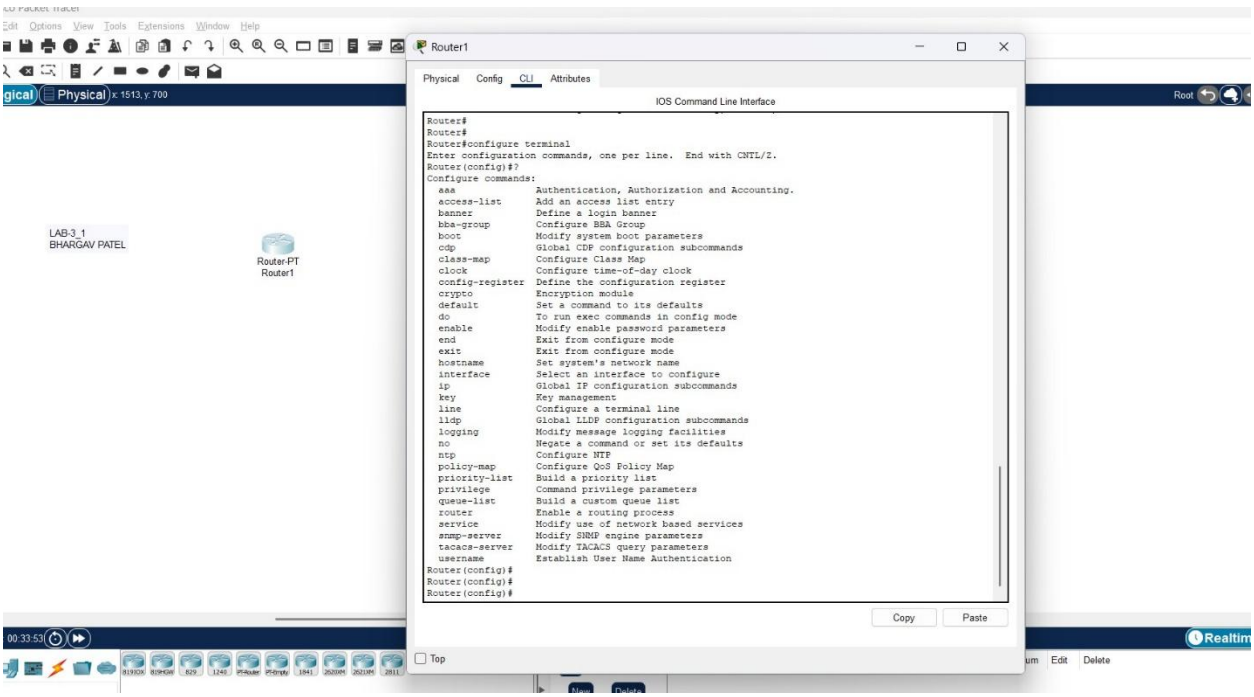
LAB-3_1
BHARGAV PATEL





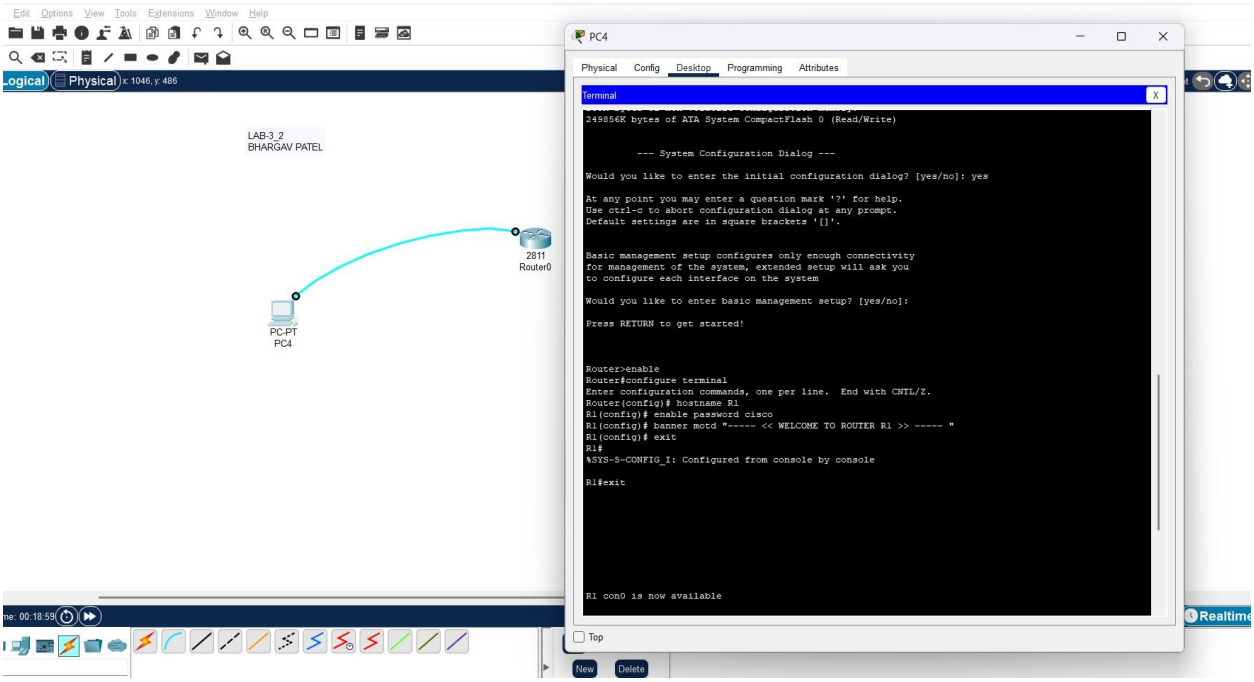
LAB-3

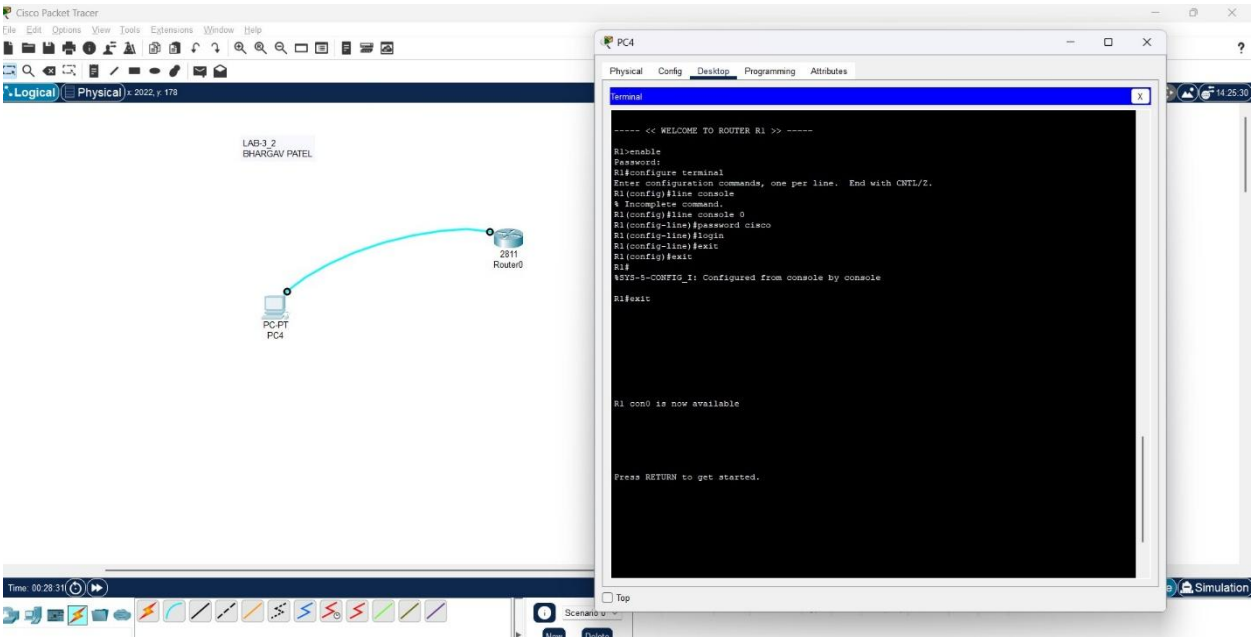

Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router#disable
Router>enable
Router#?
Exec commands:
<1-99> Session number to resume
auto Exec level Automation
clear Reset functions
clock Manage the system clock
configure Enter configuration mode
connect Open a terminal connection
copy Copy from one file to another
debug Debugging functions (see also 'undebug')
delete Delete a file
dir List files on a filesystem
disable Turn off privileged commands
disconnect Disconnect an existing network connection
enable Turn on privileged commands
erase Erase a filesystem
exit Exit from the EXEC
logout Exit from the EXEC
mkdir Create new directory
more Display the contents of a file
no Disable debugging informations
ping Send echo messages
reload Halt and perform a cold restart
resume Resume an active network connection
rmdir Remove existing directory
send Send a message to other tty lines
setup Run the SETUP command facility
show Show running system information
ssh Open a secure shell client connection
telnet Open a telnet connection
terminal Set terminal line parameters
traceroute Trace route to destination
undebug Disable debugging functions (see also 'debug')
write Write running configuration to memory, network, or terminal
Router#
Router#
Router#
Copy
Paste
Top
LAB-3_1
BHARGAV PATEL
Router-PT
Router1


Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#?
Configure commands:
aaa Authentication, Authorization and Accounting.
access-list Add an access list entry
banner Define a login banner
bba-group Configure BBA Group
boot Modify system boot parameters
cdp Global CDP configuration subcommands
class-map Configure Class Map
clock Configure time-of-day clock
config-register Define the configuration register
crypto Encryption module
default Set a command to its defaults
do To run exec commands in config mode
enable Modify enable password parameters
end Exit from configure mode
exit Exit from configure mode
hostname Set system's network name
interface Select an interface to configure
ip Global IP configuration subcommands
key Key management
line Configure a terminal line
lldp Global LLDP configuration subcommands
logging Modify message logging facilities
no Negate a command or set its defaults
ntp Configure NTP
policy-map Configure QoS Policy Map
priority-list Build a priority list
privilege Command privilege parameters
queue-list Build a custom queue list
router Enable a routing process
service Modify use of network based services
snmp-server Modify SNMP engine parameters
tacacs-server Modify TACACS query parameters
username Establish User Name Authentication
Router(config)#
Router(config)#
Router(config)#
Copy
Paste
Top
LAB-3_1
BHARGAV PATEL
Router-PT
Router1

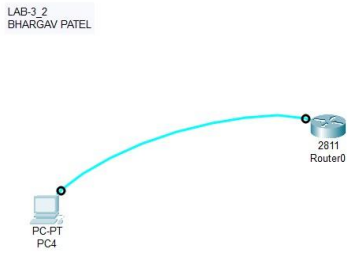
LAB-3

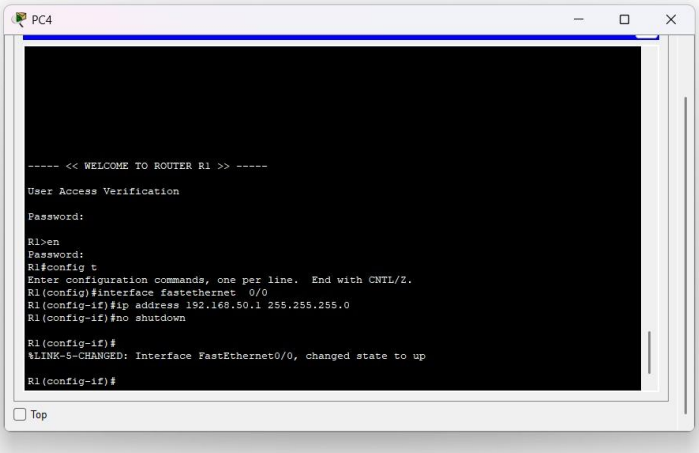
❖ LAB 3_2

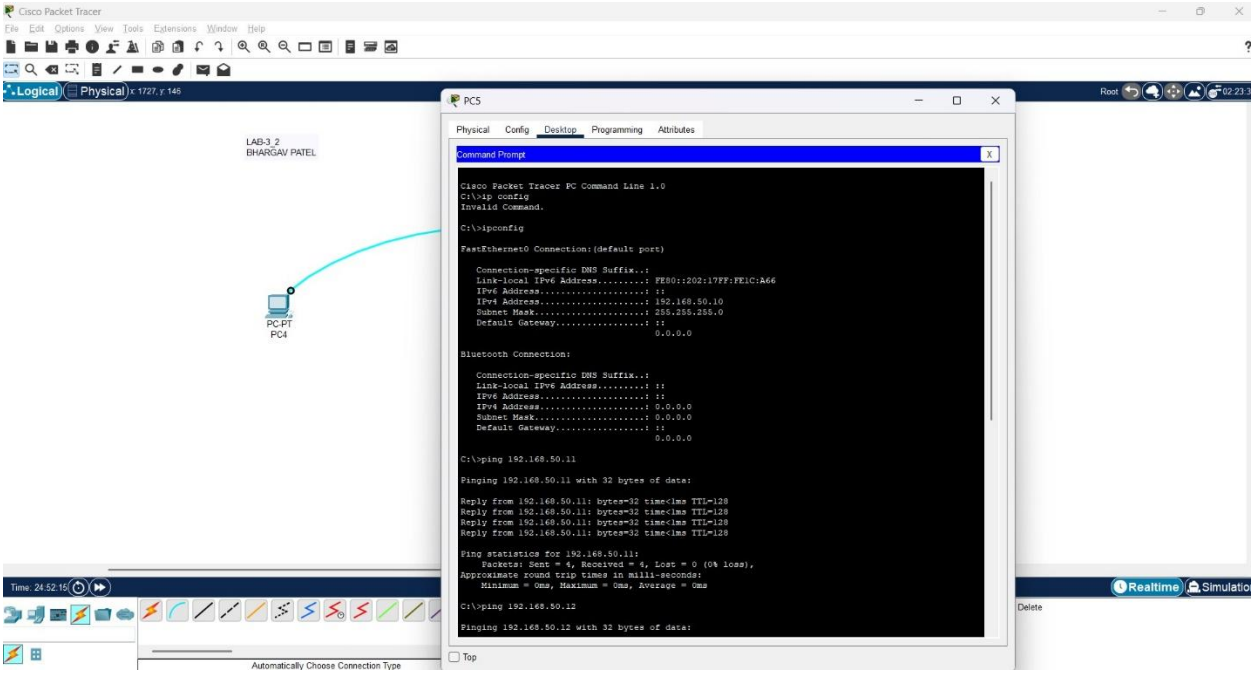

LAB-3_2
BHARGAV PATEL
2811
Router0
PC-PT
PC4
PC4
Physical
Config
Desktop
Programming
Attributes
Terminal
249856K bytes of ATA System CompactFlash 0 (Read/Write)
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: yes
At any point you may enter a question mark '?' for help.
Use ctrl-c to abort configuration dialog at any prompt.
Default settings are in square brackets '[]'.
Basic management setup configures only enough connectivity
for management of the system, extended setup will ask you
to configure each interface on the system
Would you like to enter basic management setup? [yes/no]:
Press RETURN to get started!
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# hostname R1
R1(config)# enable password cisco
R1(config)# banner motd "----- << WELCOME TO ROUTER R1 >> ----- "
R1(config)# exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
R1#exit
R1 con0 is now available
Top
New
Delete


Cisco Packet Tracer
LAB-3_2
BHARGAV PATEL
2811
Router0
PC-PT
PC4
PC4
Physical
Config
Desktop
Programming
Attributes
Terminal
----- << WELCOME TO ROUTER R1 >> -----
R1>enable
Password:
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#line console
% Incomplete command.
R1(config)#line console 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
R1#exit
R1 con0 is now available
Press RETURN to get started.
Top

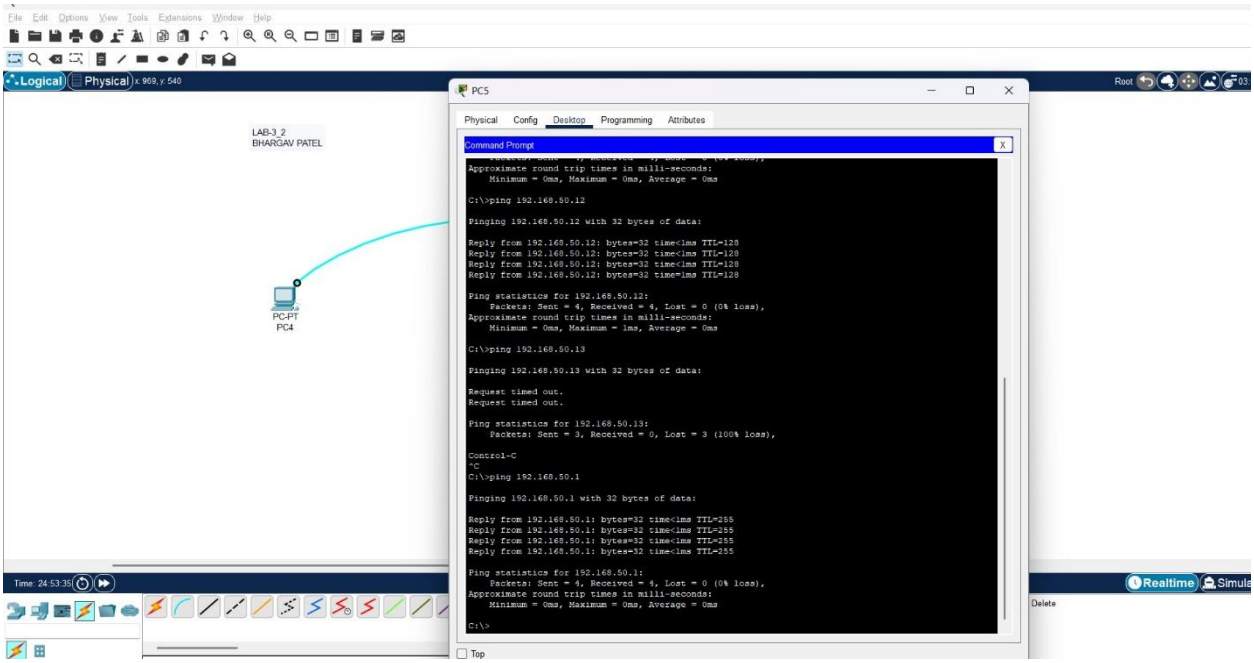
LAB-3

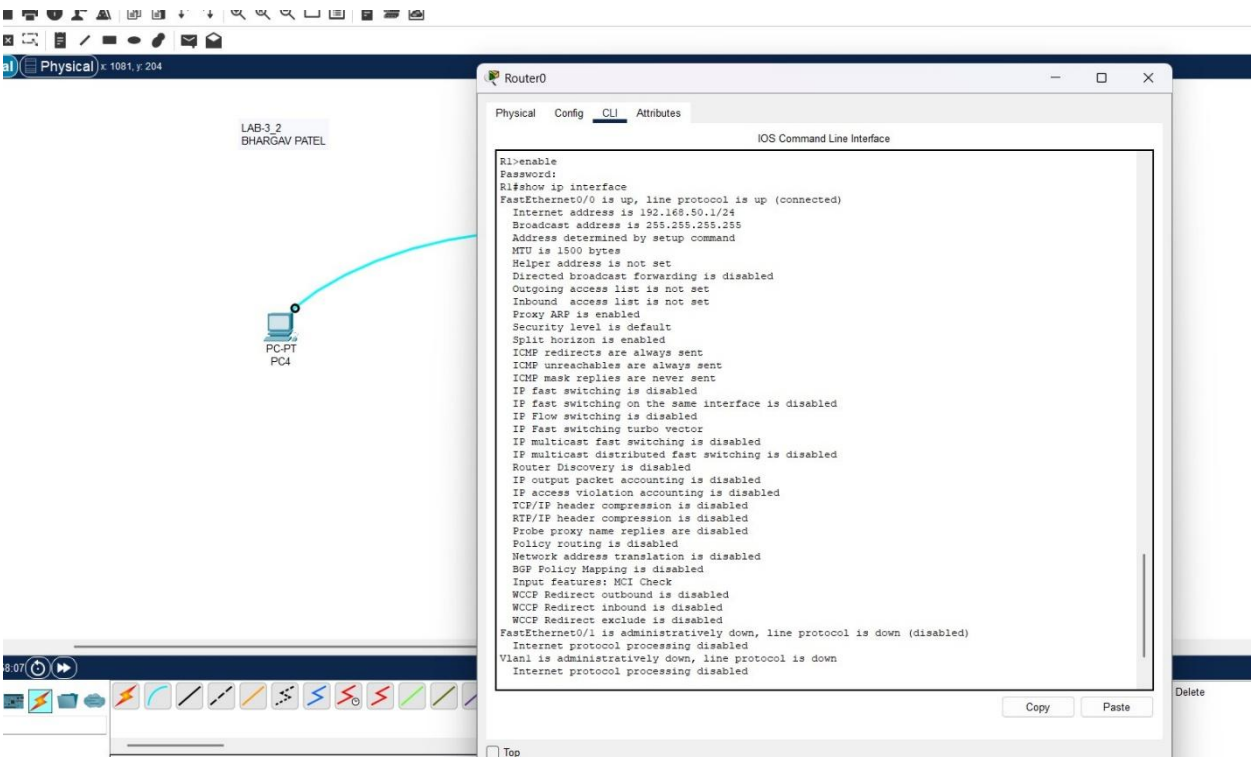

LAB-3_2
BHARGAV PATEL
2811
Router0
PC-PT
PC4


PC4
----- << WELCOME TO ROUTER R1 >> -----
User Access Verification
Password:
R1>en
Password:
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface fastethernet 0/0
R1(config-if)#ip address 192.168.50.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
R1(config-if)#
Top

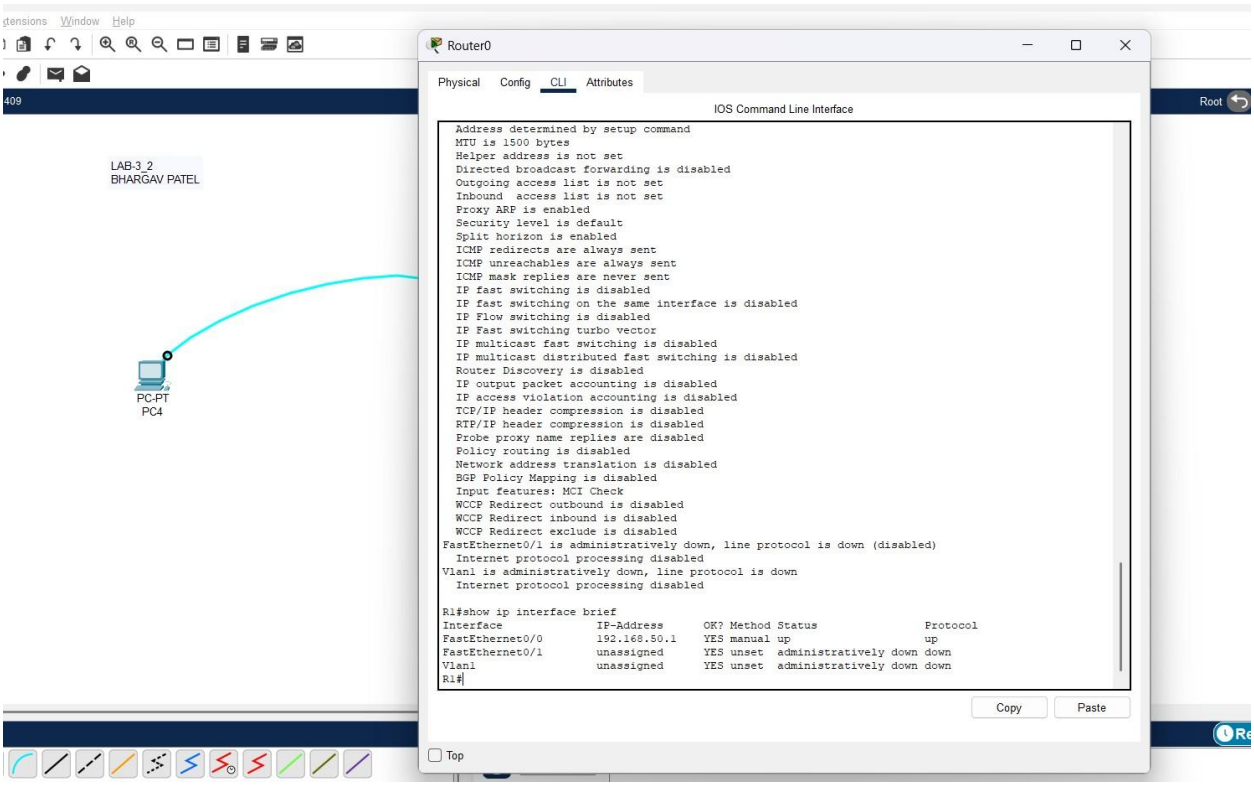

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical
LAB-3_2
BHARGAV PATEL
PC-PT
PC4
PC5
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ip config
Invalid Command.
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::202:17FF:FE1C:A66
IPv6 Address.....: ::
IPv4 Address.....: 192.168.50.10
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
0.0.0.0
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
C:\>ping 192.168.50.11
Pinging 192.168.50.11 with 32 bytes of data:
Reply from 192.168.50.11: bytes=32 time<1ms TTL=128
Reply from 192.168.50.11: bytes=32 time<1ms TTL=128
Reply from 192.168.50.11: bytes=32 time<1ms TTL=128
Reply from 192.168.50.11: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.50.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.50.12
Pinging 192.168.50.12 with 32 bytes of data:
Top
Realtime Simulation
Delete
Automatically Choose Connection Type

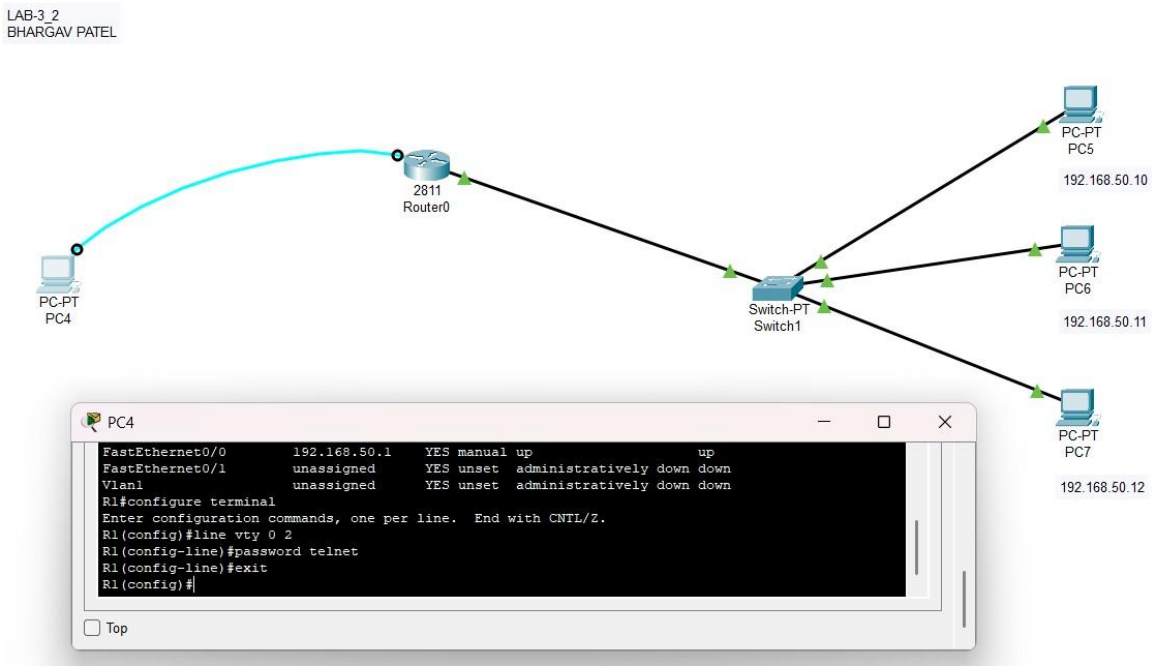
LAB-3


LAB-3_2
BHARGAV PATEL
PC-PT
PC4
PC5
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.50.12
Pinging 192.168.50.12 with 32 bytes of data:
Reply from 192.168.50.12: bytes=32 time<1ms TTL=128
Reply from 192.168.50.12: bytes=32 time<1ms TTL=128
Reply from 192.168.50.12: bytes=32 time<1ms TTL=128
Reply from 192.168.50.12: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.50.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.50.13
Pinging 192.168.50.13 with 32 bytes of data:
Request timed out.
Request timed out.
Ping statistics for 192.168.50.13:
Packets: Sent = 3, Received = 0, Lost = 3 (100% loss),
Control-C
^C
C:\>ping 192.168.50.1
Pinging 192.168.50.1 with 32 bytes of data:
Reply from 192.168.50.1: bytes=32 time<1ms TTL=255
Reply from 192.168.50.1: bytes=32 time<1ms TTL=255
Reply from 192.168.50.1: bytes=32 time<1ms TTL=255
Reply from 192.168.50.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.50.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>

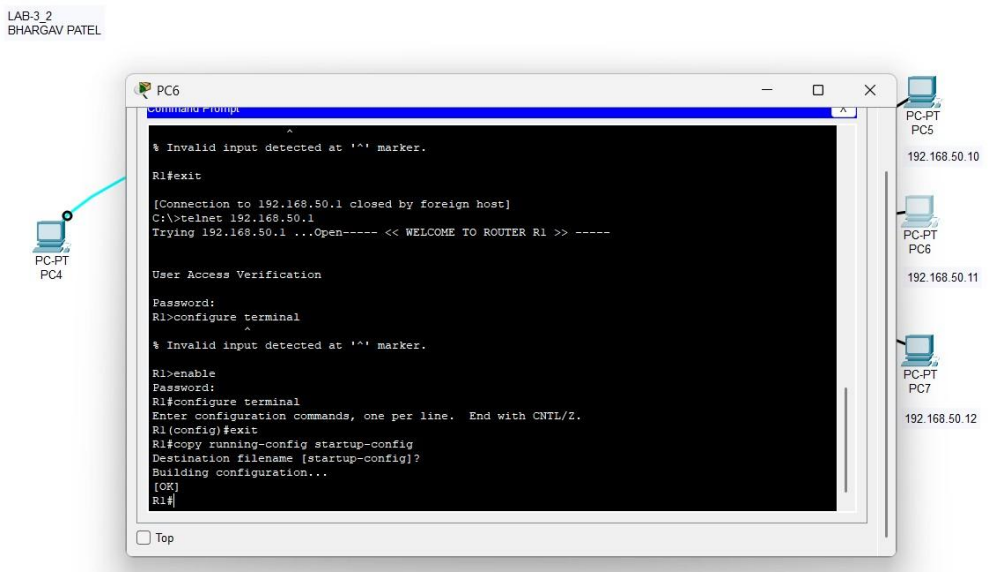

LAB-3_2
BHARGAV PATEL
PC-PT
PC4
Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
R1>enable
Password:
R1#show ip interface
FastEthernet0/0 is up, line protocol is up (connected)
Internet address is 192.168.50.1/24
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500 bytes
Helper address is not set
Directed broadcast forwarding is disabled
Outgoing access list is not set
Inbound access list is not set
Proxy ARP is enabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachables are always sent
ICMP mask replies are never sent
IP fast switching is disabled
IP fast switching on the same interface is disabled
IP Flow switching is disabled
IP Fast switching turbo vector
IP multicast fast switching is disabled
IP multicast distributed fast switching is disabled
Router Discovery is disabled
IP output packet accounting is disabled
IP access violation accounting is disabled
TCP/IP header compression is disabled
RTP/IP header compression is disabled
Probe proxy name replies are disabled
Policy routing is disabled
Network address translation is disabled
BGP Policy Mapping is disabled
Input features: MCI Check
WCCP Redirect outbound is disabled
WCCP Redirect inbound is disabled
WCCP Redirect exclude is disabled
FastEthernet0/1 is administratively down, line protocol is down (disabled)
Internet protocol processing disabled
Vlan1 is administratively down, line protocol is down
Internet protocol processing disabled
Copy
Paste

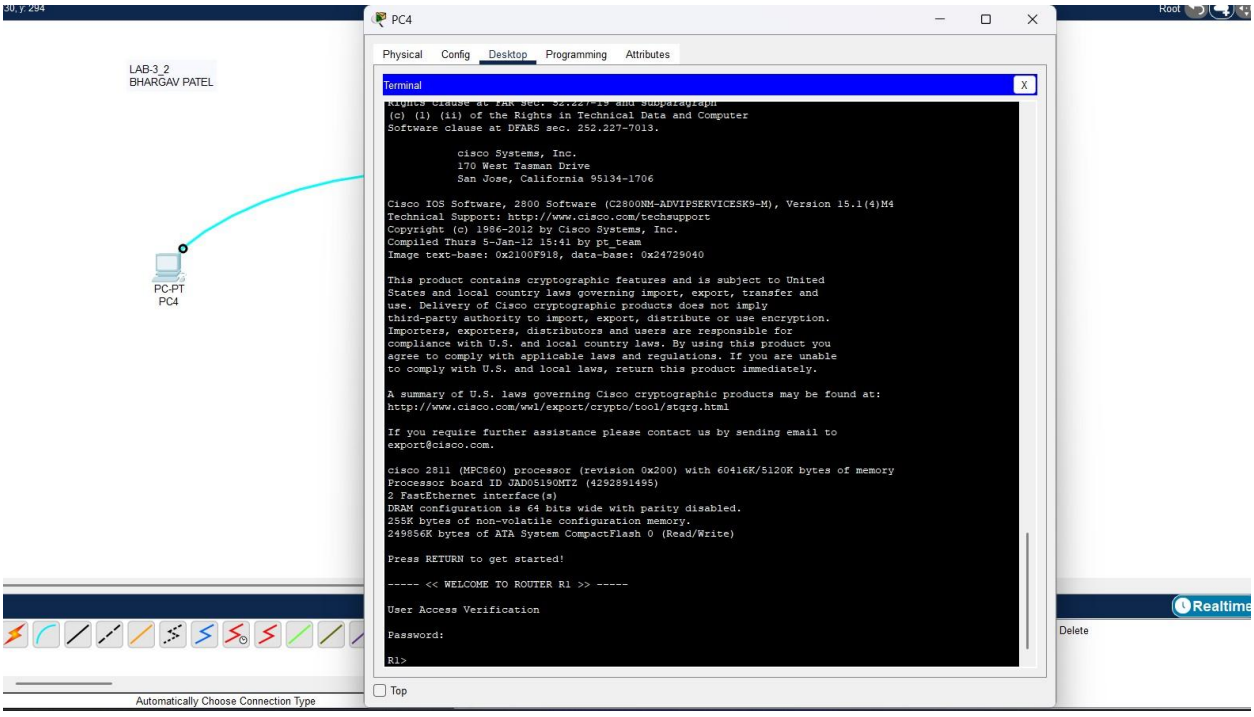
LAB-3


Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Address determined by setup command
MTU is 1500 bytes
Helper address is not set
Directed broadcast forwarding is disabled
Outgoing access list is not set
Inbound access list is not set
Proxy ARP is enabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachables are always sent
ICMP mask replies are never sent
IP fast switching is disabled
IP fast switching on the same interface is disabled
IP Flow switching is disabled
IP Fast switching turbo vector
IP multicast fast switching is disabled
IP multicast distributed fast switching is disabled
Router Discovery is disabled
IP output packet accounting is disabled
IP access violation accounting is disabled
TCP/IP header compression is disabled
RTP/IP header compression is disabled
Probe proxy name replies are disabled
Policy routing is disabled
Network address translation is disabled
BGP Policy Mapping is disabled
Input features: MCI Check
WCCP Redirect outbound is disabled
WCCP Redirect inbound is disabled
WCCP Redirect exclude is disabled
FastEthernet0/1 is administratively down, line protocol is down (disabled)
Internet protocol processing disabled
Vlan1 is administratively down, line protocol is down
Internet protocol processing disabled
R1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 192.168.50.1 YES manual up up
FastEthernet0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
R1#
Copy
Paste
Top
LAB-3_2
BHARGAV PATEL
PC-PT
PC4
Root


LAB-3_2
BHARGAV PATEL
PC-PT
PC4
2811
Router0
Switch-PT
Switch1
PC-PT
PC5
192.168.50.10
PC-PT
PC6
192.168.50.11
PC-PT
PC7
192.168.50.12
PC4
FastEthernet0/0 192.168.50.1 YES manual up up
FastEthernet0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#line vty 0 2
R1(config-line)#password telnet
R1(config-line)#exit
R1(config)#
Top

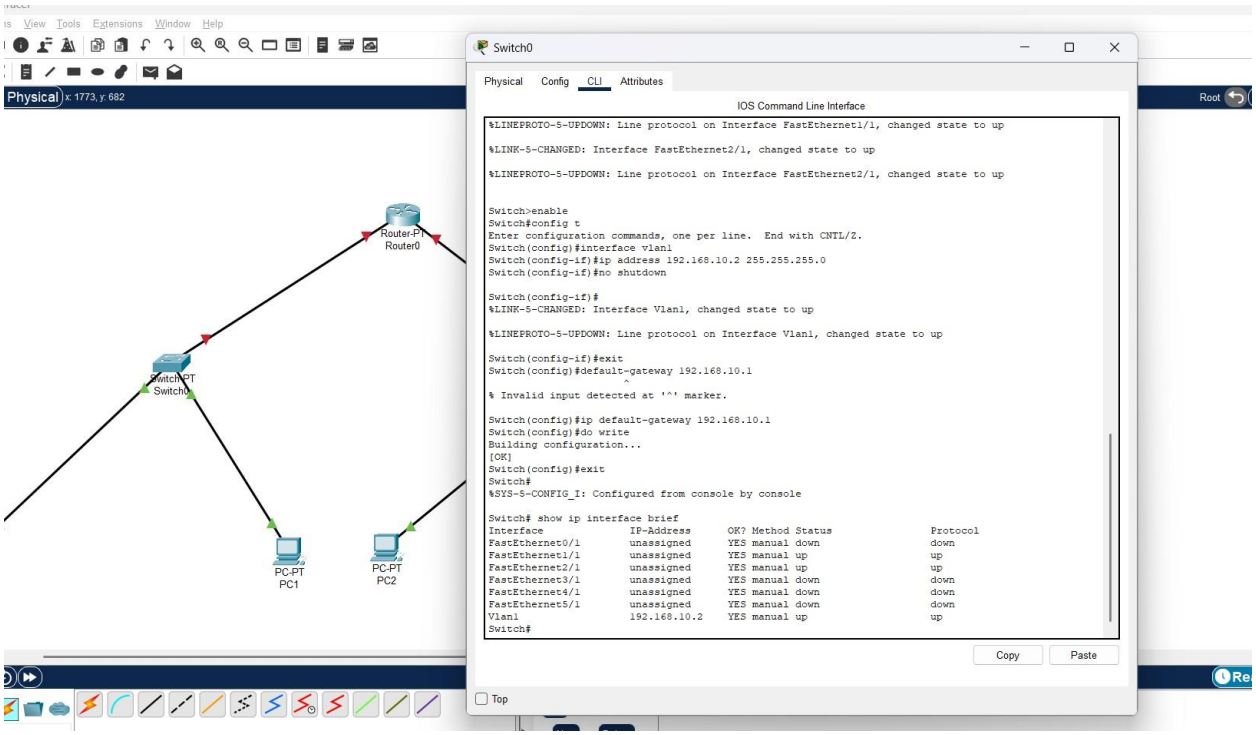
LAB-3

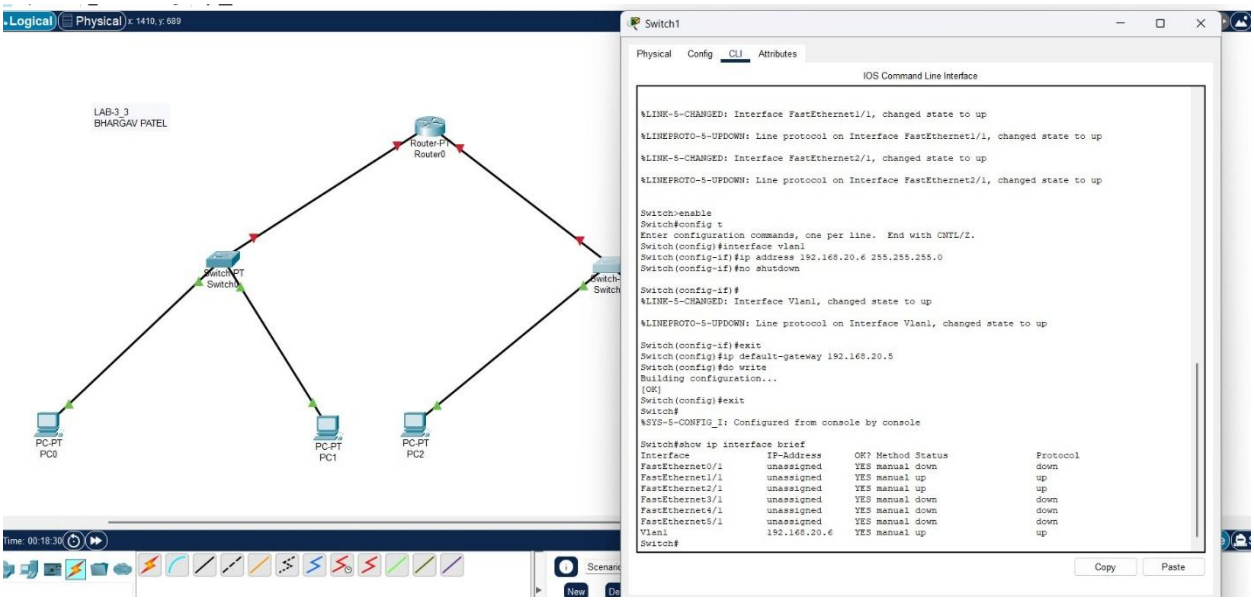

LAB-3_2
BHARGAV PATEL
PC6
% Invalid input detected at '^' marker.
R1#exit
[Connection to 192.168.50.1 closed by foreign host]
C:\>telnet 192.168.50.1
Trying 192.168.50.1 ...Open----- << WELCOME TO ROUTER R1 >> -----
User Access Verification
Password:
R1>configure terminal
% Invalid input detected at '^' marker.
R1>enable
Password:
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#exit
R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
Top
PC-PT PC4
PC-PT PC5
192.168.50.10
PC-PT PC6
192.168.50.11
PC-PT PC7
192.168.50.12


LAB-3_2
BHARGAV PATEL
PC4
Physical
Config
Desktop
Programming
Attributes
Terminal
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third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
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A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
Press RETURN to get started!
----- << WELCOME TO ROUTER R1 >> -----
User Access Verification
Password:
R1>
Top
PC-PT PC4
Realtime
Delete
Automatically Choose Connection Type

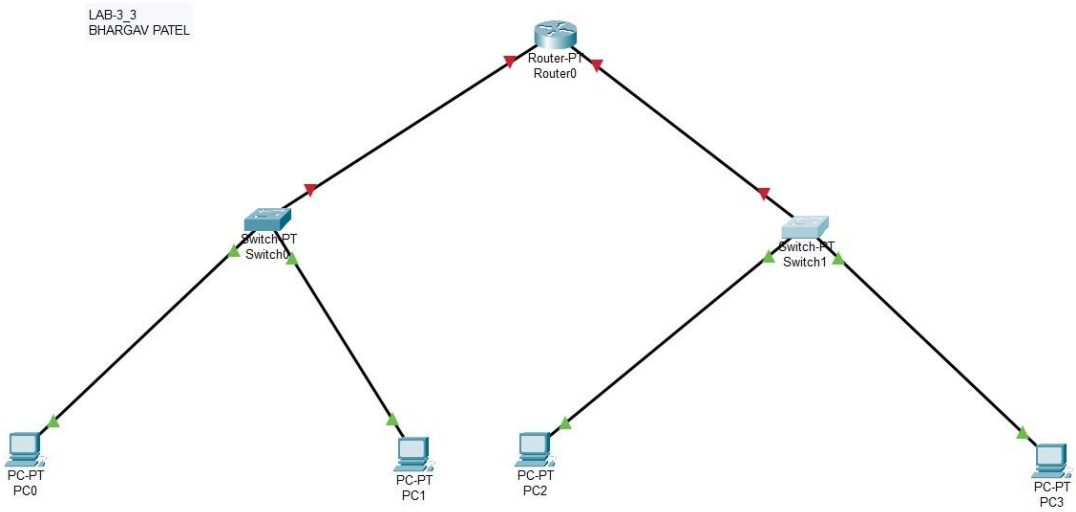
LAB-3

❖ LAB 3_3


Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet2/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan1
Switch(config-if)#ip address 192.168.10.2 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
Switch(config-if)#exit
Switch(config)#default-gateway 192.168.10.1
% Invalid input detected at '^' marker.
Switch(config)#ip default-gateway 192.168.10.1
Switch(config)#do write
Building configuration...
[OK]
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch# show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/1 unassigned YES manual down down
FastEthernet1/1 unassigned YES manual up up
FastEthernet2/1 unassigned YES manual up up
FastEthernet3/1 unassigned YES manual down down
FastEthernet4/1 unassigned YES manual down down
FastEthernet5/1 unassigned YES manual down down
Vlan1 192.168.10.2 YES manual up up
Switch#
Copy
Paste
Top
Router-PT
Router0
PC-PT
PC1
PC-PT
PC2


Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
LAB-3_3
BHARGAV PATEL
%LINK-5-CHANGED: Interface FastEthernet1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet2/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan1
Switch(config-if)#ip address 192.168.20.6 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
Switch(config-if)#exit
Switch(config)#ip default-gateway 192.168.20.5
Switch(config)#do write
Building configuration...
[OK]
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/1 unassigned YES manual down down
FastEthernet1/1 unassigned YES manual up up
FastEthernet2/1 unassigned YES manual up up
FastEthernet3/1 unassigned YES manual down down
FastEthernet4/1 unassigned YES manual down down
FastEthernet5/1 unassigned YES manual down down
Vlan1 192.168.20.6 YES manual up up
Switch#
Copy
Paste
Router-PT
Router0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2

LAB-3


LAB-3_3
BHARGAV PATEL
Router-PT
Router0
Switch-PT
Switch0
Switch-PT
Switch1
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3